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FUEL CELL COOLING SYSTEM

Abstract

In the fuel cell cooling system, a first cooling channel, a fuel cell, a second cooling channel, a radiator, a heater, a heater core, and an intermediate heat exchanger are provided, and the second cooling channel includes a bypass channel provided in parallel to the radiator, and during power generation of the fuel cell, the coolant is circulated in the first cooling channel on a path through the fuel cell and the intermediate heat exchanger, and the coolant is circulated in the second cooling channel on a path through the intermediate heat exchanger, the heater, the heater core, and the bypass channel, thereby performing a first operation of heating air by the heater core. In the first operation, the temperature of the coolant in the second cooling channel is unlikely to decrease.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-032834 filed on Mar. 5, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The technology disclosed in the present specification relates to a fuel cell cooling system.

2. Description of Related Art

[0003] A fuel cell system disclosed in Japanese Unexamined Patent Application Publication No. 2002-33108 (JP 2002-33108 A) has a first cooling channel and a second cooling channel for circulating a coolant. A fuel cell is provided on the first cooling channel. A radiator is provided on the second cooling channel. An intermediate heat exchanger is provided across the first cooling channel and the second cooling channel. The intermediate heat exchanger exchanges heat between the first cooling channel and the second cooling channel.

SUMMARY

[0004] In some cases, the fuel cell system is provided with a heater that heats the coolant and a heater core that heats air using heat of the coolant. Heating operations can be performed by the heater core. In the present specification, technology for reducing power consumption of a heater for heating coolant in a fuel cell system capable of heating operations is proposed.

First Aspect

[0005] A fuel cell cooling system disclosed in the present specification includes [0006] a first cooling channel in which coolant circulates, [0007] a fuel cell provided on the first cooling channel, [0008] a second cooling channel in which coolant circulates, [0009] a radiator provided on the second cooling channel, [0010] a heater that heats coolant in the second cooling channel, [0011] a heater core that heats air by heat exchange with coolant in the second cooling channel, and an intermediate heat exchanger that performs heat exchange between the first cooling channel and the second cooling channel, in which [0012] the second cooling channel includes a bypass channel provided in parallel to the radiator, and [0013] during power generation by the fuel cell, a first operation is executable in which coolant is circulated in the first cooling channel over a path through the fuel cell and the intermediate heat exchanger, and also coolant is circulated in the second cooling channel over a path through the intermediate heat exchanger, the heater, the heater core, and the bypass channel, and the air is heated by the heater core.

[0014] In the fuel cell cooling system described above, the first operation can be executed during power generation by the fuel cell. In the first operation, the coolant circulates in the second cooling channel over a path passing through the intermediate heat exchanger, the heater, the heater core, and the bypass channel. The coolant circulates along the path bypassing the radiator, and accordingly the temperature of the coolant in the second cooling channel does not readily decrease. Further, the heat generated in the fuel cell is transferred to the second cooling channel through the first cooling channel and the heat exchanger, and accordingly the coolant in the second cooling channel can be heated by the intermediate heat exchanger. Thus, the electric power for the heater to heat the coolant flowing through the second cooling channel is reduced. As a result, power consumption of the heater can be reduced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] Features, advantages, and technical and industrial significance of exemplary embodiments

of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0016] FIG. 1 is a block diagram of a fuel cell cooling system;

[0017] FIG. 2 is a flowchart of an operation selection process of the fuel cell cooling system;

[0018] FIG. 3 is an explanatory diagram of a path;

[0019] FIG. 4 is an explanatory diagram of a path;

[0020] FIG. 5 is an illustration of a path; and

[0021] FIG. 6 is an explanatory diagram of a path.

DETAILED DESCRIPTION OF EMBODIMENTS

[0022] Following the above aspect 1, additional configurations of the fuel cell cooling system disclosed herein are described below.

Embodiment 2

[0023] The fuel cell cooling system according to aspect 1, [0024] The second cooling channel, [0025] A first circulation system including the bypass channel, [0026] The second circulation system, [0027] A first connecting channel connecting the first circulation system and the second circulation system, [0028] A second connecting channel connecting the first circulation system and the second circulation system, [0029] The intermediate heat exchanger and the radiator is provided in the first circulation system, the heater and the heater core are provided in the second circulation system, can perform the first operation and the second operation, [0030] In the first operation, the intermediate heat exchanger, the first connecting channel, the heater, the heater core, the second connecting channel and circulating the coolant to the second cooling channel in a path passing through the bypass channel, [0031] In the second operation, during power generation of the fuel cell, the coolant is circulated in the first cooling channel through the fuel cell and the intermediate heat exchanger, the coolant is circulated in the first circulation system in a path passing through the intermediate heat exchanger and the bypass channel, and the coolant is circulated in the second circulation system in a path passing through the heater and the heater core to heat the air by the heater core.

Embodiment 3

[0032] The fuel cell cooling system according to aspect 1 or 2, [0033] During power generation of the fuel cell, a third operation of circulating a coolant in the first cooling channel on a path through the fuel cell and the intermediate heat exchanger, circulating a coolant in the second cooling channel on a path through the intermediate heat exchanger, the heater, the heater core, and the radiator, and heating the air by the heater core may be performed.

Embodiment 4

[0034] The fuel cell cooling system according to any one of aspects 1 to 3, [0035] During power generation of the fuel cell, a fourth operation can be performed in which a coolant is circulated in the first cooling channel through the fuel cell and the intermediate heat exchanger, a coolant is circulated in the first circulation system in a path through the intermediate heat exchanger and the radiator, and a coolant is circulated in the second circulation system in a path through the heater and the heater core to heat air by the heater core.

Embodiment 5

[0036] The fuel cell cooling system according to any one of aspects 1 to 4, wherein the first operation is performed when the temperature of the coolant in the first cooling channel is lower than the first reference value and the temperature of the coolant in the second cooling channel is higher than the second reference value.

Embodiment 6

[0037] The fuel cell cooling system according to any one of aspects 1 to 5, wherein the second operation is performed when the temperature of the coolant in the first cooling channel is lower than the first reference value and the temperature of the coolant in the second cooling channel is lower than the second reference value.

Embodiment 7

[0038] The fuel cell cooling system according to any one of aspects 1 to 6, wherein the third operation is performed when the temperature of the coolant in the first cooling channel is higher than the first reference value and the temperature of the coolant in the second cooling channel is higher than the second reference value.

Embodiment 8

[0039] The fuel cell cooling system according to any one of aspects 1 to 7, wherein the fourth operation is performed when the temperature of the coolant in the first cooling channel is higher than the first reference value and the temperature of the coolant in the second cooling channel is lower than the second reference value.

[0040] According to the second aspect, the temperature of the fuel cell can be increased, and the heating operation can be performed independently in the second circulation system.

[0041] According to the third aspect, since the cooling of the coolant is performed by the radiator, the cooling of the fuel cell can be efficiently performed.

[0042] According to the fourth aspect, since the cooling of the coolant is performed by the radiator, the cooling of the fuel cell can be efficiently performed. Further, since the coolant circulates independently in the second circulation system, the coolant cooled by the radiator does not flow into the second circulation system, and the power consumption of the heater can be effectively reduced.

[0043] The fuel cell cooling system **100** shown in FIG. **1** is mounted on a device (for example, a fuel cell electric vehicle) using a fuel cell as a power source. The fuel cell cooling system **100** includes a first cooling channel **12**, a second cooling channel **22**, and an intermediate heat exchanger **30**.

[0044] The intermediate heat exchanger **30** has a first heat exchange channel **30a** and a second heat exchange channel **30b**. The first heat exchange channel **30a** is a part of the first cooling channel **12**, and the second heat exchange channel **30b** is a part of the second cooling channel **22**. The intermediate heat exchanger **30** causes heat exchange between the first heat exchange channel **30a** and the second heat exchange channel **30b**. That is, the intermediate heat exchanger **30** causes heat exchange between the first cooling channel **12** and the second cooling channel **22**.

[0045] A fuel cell **14** is provided in the first cooling channel **12**. The fuel cell **14** is supplied with compressed air from an air compressor (not shown) and hydrogen from a tank (not shown). The fuel cell **14** generates electric power by reacting oxygen with hydrogen, and supplies electric power to a motor (not shown). The fuel cell **14** is cooled by the coolant in the first cooling channel **12**.

[0046] The first cooling channel **12** is an annular channel. The first cooling channel **12** is provided with a pump **16** and a temperature sensor **18a**, **18b**. The pump **16** delivers the coolant from the installation position toward the intermediate heat exchanger **30**. When the pump **16** is operated, the coolant circulates in the first cooling channel **12**.

[0047] The temperature sensor **18a** is provided in the first cooling channel **12** upstream of the fuel cell **14**. The temperature sensor **18a** detects the temperature of the coolant flowing into the fuel cell **14**. The temperature sensor **18b** is provided downstream of the fuel cell **14** in the first cooling channel **12**. The temperature sensor **18b** detects the temperature of the coolant discharged from the fuel cell **14** (hereinafter referred to as FC outlet temperature).

[0048] The second cooling channel **22** has a circulation system **61**, a circulation system **62**, a first connecting channel **34a**, and a second connecting channel **34b**. The first connecting channel **34a** connects the circulation system **61** and the circulation system **62**. The second connecting channel **34b** connects the circulation system **61** and the circulation system **62**.

[0049] The circulation system **61** includes a channel **22a**, **22b**, **22c**, a bypass channel **36**, a three-way valve **38**, and a radiator **24**. The channel **22a** is connected to the upstream end of the second heat exchange channel **30b** and the downstream end of the coolant channel in the radiator **24**. The channel **22b** is connected to the three-way valve **38** and the downstream end of the second heat

exchange channel **30b**. The channel **22c** is connected to the three-way valve **38** and the upstream end of the coolant channel in the radiator **24**. An upstream end of the bypass channel **36** is connected to the three-way valve **38**. The downstream end of the bypass channel **36** is connected to the channel **22a**. The three-way valve **38** switches between a state in which the channel **22b** is connected to the channel **22c** and a state in which the channel **22b** is connected to the bypass channel **36**. The radiator **24** cools the coolant flowing in the circulation system **61** by heat exchange with the outside air.

[0050] The circulation system **61** is provided with a pump **26** and a temperature-sensor **28a**, **28b**, **28c**. The pump **26** is a channel **22a** and is provided between the intermediate heat exchanger **30** and the bypass channel **36**. The pump **26** delivers the coolant from the installation position toward the intermediate heat exchanger **30**. When the pump **26** is operated, the coolant circulates in the second cooling channel **22**.

[0051] The temperature sensor **28a** is provided in the channel **22a**. The temperature sensor **28a** detects the temperature of the coolant emitted from the radiator **24**. The temperature sensor **28b** is provided in the channel **22c**. The temperature sensor **28b** detects the temperature of the coolant flowing into the radiator **24**. The temperature sensor **28c** is provided in the channel **22b**. The temperature sensor **28c** detects the temperature of the coolant emitted from the intermediate heat exchanger **30**. That is, the temperature sensor **28c** detects the outlet temperature of the intermediate heat exchanger **30**.

[0052] The circulation system **62** has an air-conditioning channel **50**. The air-conditioning channel **50** is an annular channel. The first connecting channel **34a** connects the air-conditioning channel **50** and the channel **22b** of the circulation system **61**. The second connecting channel **34b** connects the air-conditioning channel **50** and the channel **22b** of the circulation system **61**. The connecting portion between the first connecting channel **34a** and the channel **22b** is located upstream of the connecting portion between the second connecting channel **34b** and the channel **22b** in the channel **22b**. The air-conditioning channel **50** is provided with a heater **52**, a heater core **54**, a pump **56**, and a three-way valve **58**.

[0053] The three-way valve **58** is provided at a connection portion between the first connecting channel **34a** and the air-conditioning channel **50**. In other words, the three-way valve **58** is connected to the first connecting channel **34a**, the upstream portion of the air-conditioning channel **50**, and the downstream portion of the air-conditioning channel **50**. The three-way valve **58** switches between a first state in which the coolant flows from the first connecting channel **34a** to the air-conditioning channel **50** and a second state in which the coolant does not flow from the first connecting channel **34a** to the air-conditioning channel **50**. In the first condition, the coolant flows from the first connecting channel **34a** to the downstream portion of the air-conditioning channel **50** via the three-way valve **58**. In the second state, the coolant flows from the upstream portion of the air-conditioning channel **50** to the downstream portion of the air-conditioning channel **50** via the three-way valve **58**.

[0054] The pump **56** is provided in the air-conditioning channel **50** on the downstream side of the three-way valve **58**. The pump **56** delivers the coolant from the installation position toward the heater **52**.

[0055] The heater **52** is provided in the air-conditioning channel **50** on the downstream side of the pump **56**. The heater **52** heats the coolant flowing through the air-conditioning channel **50**.

[0056] The heater core **54** is provided in the air-conditioning channel **50** on the downstream side of the heater **52**. The heater core **54** performs heat exchange between the coolant in the air-conditioning channel **50** and the outside air, and heats the air. The air heated by the heater core **54** is supplied to the vehicle interior, thereby adjusting the vehicle interior temperature.

[0057] When the fuel cell **14** generates electricity, the fuel cell **14** generates heat. The control device **40** activates the pumps **16**, **26** during power generation of the fuel cell **14**. When the pump **16** is operated, the coolant circulates in the first cooling channel **12**. When the pump **26** is operated,

the coolant circulates in the second cooling channel **22**. The fuel cell **14** is cooled by heat exchange between the coolant in the first cooling channel **12** and the fuel cell **14**. The heat generated by the power generation of the fuel cell **14** is transferred to the intermediate heat exchanger **30** (that is, the first heat exchange channel **30a**) by the coolant in the first cooling channel **12**. The intermediate heat exchanger **30** transfers heat from the first heat exchange channel **30a** to the second heat exchange channel **30b**.

[0058] The fuel cell cooling system **100** executes the flowchart shown in FIG. 2 during power generation of the fuel cell **14**.

[0059] In **S2**, the control device **40** determines whether FC outlet temperature is less than the determination temperature **T1**. If FC outlet temperature is less than the determination temperature **T1** (if **S2** is YES), the control device **40** controls the three-way valve **38** with the bypass channel **36** open (**S4**). In this case, no coolant flows through the radiator **24**. In addition, when FC outlet temperature is equal to or higher than the determination temperature **T1** (NO in **S2**), the control device **40** controls the three-way valve **38** with the bypass channel **36** closed (**S6**). In this case, no coolant flows through the bypass channel **36**.

[0060] Next, in **S8**, the control device **40** determines whether there is a heating demand. If there is no heating requirement (NO in **S8**), the control device **40** controls the three-way valve **58** with the first connecting channel **34a** closed (**S10**). In this case, the coolant does not flow through the first connecting channel **34a** and the second connecting channel **34b**. That is, the coolant does not flow between the circulation system **61** and the circulation system **62**. Also, in **S10**, the control device **40** does not activate the pump **56** and heaters **52**. Therefore, the heater core **54** does not perform heating.

[0061] When FC outlet temperature is low (i.e., when the bypass channel **36** is open), in **S10**, the coolant delivered by the pump **26** circulates in the second cooling channel **22** in a path passing through the intermediate heat exchanger **30**, the channel **22b**, and the bypass channel **36**. Since the coolant in the second cooling channel **22** is not cooled by the radiator **24**, the temperature of the coolant flowing into the intermediate heat exchanger **30** is not very low. Therefore, in the intermediate heat exchanger **30**, the cooling efficiency when cooling the coolant in the first cooling channel **12** is low. Therefore, the cooling efficiency of the fuel cell **14** is low, and the temperature of the fuel cell **14** gradually increases. In this way, when FC outlet temperature is low (that is, when the temperature of the fuel cell **14** is low), the temperature of the fuel cell **14** is increased to increase the power generation efficiency of the fuel cell **14**.

[0062] When FC outlet temperature is high (i.e., when the bypass channel **36** is closed), in **S10**, the coolant delivered by the pump **26** circulates in the second cooling channel **22** in a path passing through the intermediate heat exchanger **30**, the channel **22b**, the channel **22c**, the radiator **24**, and the channel **22a**. Since the coolant in the second cooling channel **22** is cooled by the radiator **24**, the temperature of the coolant flowing into the intermediate heat exchanger **30** is low. Therefore, in the intermediate heat exchanger **30**, the coolant in the first cooling channel **12** is efficiently cooled. Therefore, the fuel cell **14** is efficiently cooled by the coolant in the first cooling channel **12**. Thus, when FC outlet temperature is high (i.e., when the temperature of the fuel cell **14** is high), the radiator **24** is operated to efficiently cool the fuel cell **14**.

[0063] The control device **40** repeatedly executes **S10**, **S26** from **S2** while there is no heating demand. Thus, while there is no heating demand, the control device **40** performs **S10** operations described above until the power of the vehicles is OFF (i.e., YES at **S26**).

[0064] The control device **40** activates the pump **56** when there is a heating demand (**S8** and YES). The coolant sent out from the pump **56** flows in the order of the heater **52** and the heater core **54**.

[0065] After the pump **56** is activated, the control device **40** determines whether the outside air temperature is less than or equal to the cold determination temperature **T2** in **S12**. When the outside air temperature is equal to or lower than the cold determination temperature **T2** (YES in **S12**), the control device **40** next determines whether or not the outlet temperature of the intermediate heat

exchanger **30** is equal to or lower than the determination temperature T3 (S14).

[0066] When the outlet temperature of the intermediate heat exchanger **30** is equal to or lower than the determination temperature T3 (YES in S14), the control device **40** controls the three-way valve **58** with the first connecting channel **34a** closed (S16). In this case, the coolant does not flow through the first connecting channel **34a** and the second connecting channel **34b**. That is, the coolant does not flow between the circulation system **61** and the circulation system **62**.

[0067] When FC outlet temperature is low (i.e., when the bypass channel **36** is open), in S16, the coolant in the second cooling channel **22** circulates in the paths **102** and **104** indicated by the arrows in FIG. 3. In the path **102**, the coolant delivered by the pump **26** circulates in the circulation system **61** in a path passing through the intermediate heat exchanger **30**, the channel **22b**, and the bypass channel **36**. Since the coolant in the circulation system **61** is not cooled by the radiator **24**, the temperature of the coolant flowing into the intermediate heat exchanger **30** is not very low. Therefore, in the intermediate heat exchanger **30**, the cooling efficiency when cooling the coolant in the first cooling channel **12** is low. Therefore, the cooling efficiency of the fuel cell **14** is low, and the temperature of the fuel cell **14** gradually increases. As described above, when FC outlet temperature is lower, the temperature of the fuel cell **14** is increased to increase the power generation efficiency of the fuel cell **14**.

[0068] In the path **104**, the coolant delivered by the pump **56** circulates in the circulation system **62** (that is, the air-conditioning channel **50**) in a path passing through the heater **52** and the heater core **54**. The heater **52** heats the coolant flowing through the air-conditioning channel **50**. The heater core **54** performs heat exchange between the coolant in the air-conditioning channel **50** and the outside air, and heats the air. As a result, the air heated by the heater core **54** is supplied to the vehicle interior, thereby adjusting the vehicle interior temperature.

[0069] In this way, in S16 where FC outlet temperature is low, the coolant is circulated independently in the circulation system **61** (i.e., the path **102**) and the circulation system **62** (i.e., the path **104**). Therefore, the low-temperature coolant in the circulation system **61** does not flow into the circulation system **62**, and it is possible to appropriately perform heating in the heater core **54** by using the heat of the coolant heated by the heater **52**.

[0070] When FC outlet is hot (i.e., when the bypass channel **36** is closed), the coolant in the second cooling channel **22** circulates in the paths **106** and **104** indicated by the arrows in FIG. 4 in S16. The path **104** shown in FIG. 4 is the same as the path **104** shown in FIG. 3. In the path **106**, the coolant delivered by the pump **26** circulates in the circulation system **61** in a path passing through the intermediate heat exchanger **30**, the channel **22b**, the channel **22c**, the radiator **24**, and the channel **22a**. Since the coolant in the circulation system **61** is cooled by the radiator **24**, the temperature of the coolant flowing into the intermediate heat exchanger **30** is low. Therefore, in the intermediate heat exchanger **30**, the coolant in the first cooling channel **12** is efficiently cooled. Therefore, the fuel cell **14** is efficiently cooled by the coolant in the first cooling channel **12**. Thus, when FC outlet is hot, the radiator **24** is activated to efficiently cool the fuel cell **14**.

[0071] In this way, in S16 where FC outlet temperature is higher, the coolant is circulated independently in the circulation system **61** (i.e., the path **106**) and the circulation system **62** (i.e., the path **104**). Therefore, the low-temperature coolant in the circulation system **61** does not flow into the circulation system **62**, and it is possible to appropriately perform heating in the heater core **54** by using the heat of the coolant heated by the heater **52**.

[0072] The control device **40** repeatedly executes S16, S26 from S2 to S8, S12 while the outlet temperature of the intermediate heat exchanger **30** is equal to or lower than the determination temperature T3. Therefore, while the outlet temperature of the intermediate heat exchanger **30** is equal to or lower than the determination temperature T3, the control device **40** executes the above-described S16 operation until the power supply of the vehicle is turned OFF (that is, YES at S26).

[0073] If the outlet temperature of the intermediate heat exchanger **30** exceeds the determination temperature T3 (NO in S14), the control device **40** controls the three-way valve **58** with the first

connecting channel **34a** open (S18). Here, the coolant flows through the first connecting channel **34a** and the second connecting channel **34b**. That is, the coolant circulates between the circulation system **61** and the circulation system **62**.

[0074] When FC outlet temperature is low (i.e., when the bypass channel **36** is open), in S18, the coolant in the second cooling channel **22** circulates in the path **108** indicated by the arrow in FIG. 5. In the path **108**, the coolant delivered by the pump **26** circulates in the second cooling channel **22** in a path passing through the intermediate heat exchanger **30**, the channel **22b**, the first connecting channel **34a**, the heater **52**, the heater core **54**, the second connecting channel **34b**, and the bypass channel **36**. Since the coolant in the second cooling channel **22** is not cooled by the radiator **24**, the temperature of the coolant flowing into the intermediate heat exchanger **30** is not very low. Therefore, in the intermediate heat exchanger **30**, the cooling efficiency when cooling the coolant in the first cooling channel **12** is low. Therefore, the cooling efficiency of the fuel cell **14** is low, and the temperature of the fuel cell **14** gradually increases. As the temperature of the fuel cell **14** increases, the power generation efficiency of the fuel cell **14** increases.

[0075] In the intermediate heat exchanger **30**, the coolant in the second cooling channel **22** is heated by heat exchange between the first cooling channel **12** and the second cooling channel **22**. As described above, S18 is performed when the outlet temperature of the intermediate heat exchanger **30** exceeds the determination temperature T3. Therefore, in S18, the high-temperature coolant that has passed through the intermediate heat exchanger **30** flows into the air-conditioning channel **50**. The heater **52** further heats the coolant in the air-conditioning channel **50**, and the heater core **54** performs heating by using the heat of the coolant in the air-conditioning channel **50**. Since the temperature of the coolant supplied from the intermediate heat exchanger **30** to the air-conditioning channel **50** is high, it is not necessary to raise the temperature of the coolant by the heater **52** so much. Therefore, the heater **52** can be operated at a low output, and the power consumption of the heater **52** is reduced. As described above, when the outlet temperature of the intermediate heat exchanger **30** is high, the heat generated in the fuel cell **14** is supplied to the heater core **54** via the first cooling channel **12**, the intermediate heat exchanger **30**, and the second cooling channel **22**, so that the power consumption of the heater **52** can be reduced.

[0076] When FC outlet is hot (i.e., when the bypass channel **36** is closed), the coolant in the second cooling channel **22** circulates in the path **110** indicated by the arrow in FIG. 6 in S18. In the path **110**, the coolant delivered by the pump **26** circulates in the second cooling channel **22** in a path passing through the intermediate heat exchanger **30**, the channel **22b**, the first connecting channel **34a**, the heater **52**, the heater core **54**, the second connecting channel **34b**, the channel **22c**, the radiator **24**, and the channel **22a**. Since the coolant in the second cooling channel **22** is cooled by the radiator **24**, the temperature of the coolant flowing into the intermediate heat exchanger **30** is low. Therefore, in the intermediate heat exchanger **30**, the coolant in the first cooling channel **12** is efficiently cooled. Therefore, the fuel cell **14** is efficiently cooled by the coolant in the first cooling channel **12**.

[0077] In the intermediate heat exchanger **30**, the coolant in the second cooling channel **22** is heated by heat exchange between the first cooling channel **12** and the second cooling channel **22**. As described above, S18 is performed when the outlet temperature of the intermediate heat exchanger **30** exceeds the determination temperature T3. Therefore, in S18, the high-temperature coolant that has passed through the intermediate heat exchanger **30** flows into the air-conditioning channel **50**. The heater **52** further heats the coolant in the air-conditioning channel **50**, and the heater core **54** performs heating by using the heat of the coolant in the air-conditioning channel **50**. Since the temperature of the coolant supplied from the intermediate heat exchanger **30** to the air-conditioning channel **50** is high, it is not necessary to raise the temperature of the coolant by the heater **52** so much. Therefore, the heater **52** can be operated at a low output, and the power consumption of the heater **52** is reduced. As described above, when the outlet temperature of the intermediate heat exchanger **30** is high, the heat generated in the fuel cell **14** is supplied to the

heater core **54** via the first cooling channel **12**, the intermediate heat exchanger **30**, and the second cooling channel **22**, so that the power consumption of the heater **52** can be reduced.

[0078] The control device **40** repeatedly executes **S8**, **S12**, **S14**, **S18**, **S26** from **S2** while the outlet temperature of the intermediate heat exchanger **30** exceeds the determination temperature **T3**. Thus, while the outlet temperature of the intermediate heat exchanger **30** is above the determination temperature **T3**, the control device **40** performs **S18** operations described above until the power of the vehicle is turned OFF (i.e., YES at **S26**).

[0079] When the outside air temperature exceeds the cold determination temperature **T2** (when **S12** is NO), the control device **40** determines whether or not the outlet temperature of the intermediate heat exchanger **30** is equal to or lower than the determination temperature **T4** (**S20**). The determination temperature **T4** differs from the determination temperature **T3** of **S14**.

[0080] When the outlet temperature of the intermediate heat exchanger **30** is equal to or lower than the determination temperature **T4** (YES in **S20**), the control device **40** executes **S22**. In **S22**, the same operation as that of **S16** is executed.

[0081] When the outlet temperature of the intermediate heat exchanger **30** exceeds the determination temperature **T4** (NO in **S20**), the control device **40** executes **S24**. In **S24**, the same operation as that of **S18** is executed.

[0082] As described above, **S24** from **S20** is the same as that from **S14** to **S18** except that the determination times differ.

[0083] As described above, the control device **40** selectively executes the operations of FIGS. **3** to **6** when there is a heating request.

[0084] When the temperature of the coolant in the second cooling channel **22** is high, the coolant in the second cooling channel **22** is circulated in a path that straddles the circulation system **61** and the circulation system **62** by the operation of FIG. **5** or **6**. As a result, the heat generated in the fuel cell **14** can be used in the heater core **54**, and the power consumption of the heater **52** can be reduced. In this case, when the temperature of the fuel cell **14** is low, the temperature of the fuel cell **14** is increased by the operation of FIG. **5** (that is, the operation of not cooling the coolant by the radiator **24**). When the temperature of the fuel cell **14** is high, the temperature of the fuel cell **14** is lowered by the operation of FIG. **6** (that is, the operation of cooling the coolant by the radiator **24**). Therefore, it is possible to prevent the temperature of the fuel cell **14** from becoming excessively high while suppressing a decrease in power generation efficiency due to a decrease in the temperature of the fuel cell **14**.

[0085] In addition, when the temperature of the coolant in the second cooling channel **22** is low, the coolant in the second cooling channel **22** is circulated through the respective independent paths of the circulation system **61** and the circulation system **62** by the operation of FIG. **3** or **4**.

Accordingly, it is possible to prevent the low-temperature coolant from flowing into the air-conditioning channel **50**, and to prevent an increase in power consumption of the heater **52**. In this case, when the temperature of the fuel cell **14** is low, the temperature of the fuel cell **14** is increased by the operation of FIG. **3** (that is, the operation of not cooling the coolant by the radiator **24**).

When the temperature of the fuel cell **14** is high, the temperature of the fuel cell **14** is lowered by the operation of FIG. **4** (that is, the operation of cooling the coolant by the radiator **24**). Therefore, it is possible to prevent the temperature of the fuel cell **14** from becoming excessively high while suppressing a decrease in power generation efficiency due to a decrease in the temperature of the fuel cell **14**.

[0086] The operation of executing **S18** while the bypass channel **36** of the embodiment is open is an example of the first operation. The operation of executing **S16** while the bypass channel **36** of the embodiment is open is an example of the second operation. The operation of the embodiment is an example of the third operation in which **S18** is executed while the bypass channel **36** is closed. The operation of executing **S16** while the bypass channel **36** of the embodiment is closed is an example of the fourth operation.

[0087] In the embodiment, the flow path changing device for changing the flowing path of the coolant in the second cooling channel **22** is configured by the three-way valves **38** and **58**, but the flow path changing device may be configured by other devices. Further, in the embodiment, the three-way valves **38** and **58** are solenoid valves, but the channel changing device may be configured by a device that does not use electricity. For example, a channel changing device may be configured by a device that switches a channel by thermal expansion of a material, such as a thermostat.

[0088] In the embodiment, the first to fourth operations were performed by comparing the outlet temperature of the intermediate heat exchanger **30** with the determination temperature **T3**, **T4**. However, the first to fourth operations may be performed by comparing FC outlet temperature with the determination temperature **T3**, **T4**.

[0089] While the embodiments have been described in detail above, these are merely illustrative and do not limit the scope of the claims. The technology described in the claims includes various modifications and alterations of the specific examples described above. The technical elements described in this specification or in the drawings may be used alone or in various combinations, and are not limited to the combinations described in the claims at the time of filing. Further, the technology illustrated in the present specification or the drawings achieves a plurality of objects at the same time, and has technical usefulness by achieving one of the objects.

Claims

1. A fuel cell cooling system, comprising: a first cooling channel in which coolant circulates; a fuel cell provided on the first cooling channel; a second cooling channel in which coolant circulates; a radiator provided on the second cooling channel; a heater that heats coolant in the second cooling channel; a heater core that heats air by heat exchange with coolant in the second cooling channel; and an intermediate heat exchanger that performs heat exchange between the first cooling channel and the second cooling channel, wherein: the second cooling channel includes a bypass channel provided in parallel to the radiator; and during power generation by the fuel cell, a first operation is executable in which coolant is circulated in the first cooling channel over a path through the fuel cell and the intermediate heat exchanger, and also coolant is circulated in the second cooling channel over a path through the intermediate heat exchanger, the heater, the heater core, and the bypass channel, and the air is heated by the heater core.

2. The fuel cell cooling system according to claim 1, wherein: the second cooling channel includes a first circulation system including the bypass channel, a second circulation system, a first connecting channel connecting the first circulation system and the second circulation system, and a second connecting channel connecting the first circulation system and the second circulation system; the intermediate heat exchanger and the radiator are provided in the first circulation system; the heater and the heater core are provided in the second circulation system; the first operation and a second operation are executable; in the first operation, the coolant is circulated in the second cooling channel over a path passing through the intermediate heat exchanger, the first connecting channel, the heater, the heater core, the second connecting channel, and the bypass channel; and in the second operation, during power generation by the fuel cell, coolant is circulated in the first cooling channel over a path passing through the fuel cell and the intermediate heat exchanger, coolant is circulated in the first circulation system over a path passing through the intermediate heat exchanger and the bypass channel, and coolant is circulated in the second circulation system over a path passing through the heater and the heater core to heat the air by the heater core.

3. The fuel cell cooling system according to claim 1, wherein, during power generation by the fuel cell, a third operation is executable in which coolant is circulated in the first cooling channel over a path passing through the fuel cell and the intermediate heat exchanger, and also coolant is

circulated in the second cooling channel over a path passing through the intermediate heat exchanger, the heater, the heater core, and the radiator to heat air by the heater core.

4. The fuel cell cooling system according to claim 2, wherein, during power generation of the fuel cell, a fourth operation is executable in which coolant is circulated in the first cooling channel over a path passing through the fuel cell and the intermediate heat exchanger, the coolant is circulated in the first circulation system over a path passing through the intermediate heat exchanger and the radiator, and coolant is circulated in the second circulation system over a path passing through the heater and the heater core to heat air by the heater core.
